RISK MANAGEMENT PLAN

Roaya

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Smart City Transformation · Jan 2027 – Dec 2028 · Plan owner: Risk & Quality Manager

1. Risk Identification

ID	Risk	Type	Description
R1	Cyberattack on city systems	Security	Hackers attack the control center, cameras or citizen data
R2	Technical issues with smart devices	Technical	Sensors or cameras fail, or systems are not compatible
R3	Budget overrun	Financial	Costs go higher than the EGP 550M budget
R4	Data privacy concerns	Legal	Citizens worry about surveillance and personal data
R5	Delays in regulatory approvals	External	Government approvals take longer than planned
R6	Contractor or supplier delays	External	Equipment or construction work arrives late
R7	Insufficient user adoption	Organizational	Citizens don't use the new digital services

2. Risk Analysis

Score = Probability (1–5) × Impact (1–5). Risks are sorted from the highest score.

ID	Risk	Probability	Impact	Score	Severity
R1	Cyberattack on city systems	Possible (3)	Catastrophic (5)	15	High
R2	Technical issues with smart devices	Likely (4)	Moderate (3)	12	High
R3	Budget overrun	Possible (3)	Major (4)	12	High
R4	Data privacy concerns	Possible (3)	Major (4)	12	High
R6	Contractor or supplier delays	Possible (3)	Moderate (3)	9	Moderate
R7	Insufficient user adoption	Possible (3)	Moderate (3)	9	Moderate
R5	Delays in regulatory approvals	Unlikely (2)	Major (4)	8	Moderate

Risk Matrix

Impact ↓ / Probability →	Rare (1)	Unlikely (2)	Possible (3)	Likely (4)	Almost certain (5)
Catastrophic (5)			R1		
Major (4)		R5	R3 R4		
Moderate (3)			R6 R7	R2	
Minor (2)					
Insignificant (1)					

Low (1–4)
Handled by the risk owner

Moderate (5–10)
Owner acts; reviewed monthly

High (12–16)
Project Manager involved; reviewed weekly

Extreme (20–25)
Escalate to the sponsor immediately

3–4. Risk Response and Owners

Strategy: Avoid (remove the cause) · Reduce (lower probability or impact) · Transfer (share it with a contractor or insurer) · Accept (monitor only).

ID	Risk	Strategy	Action to minimize the risk	Owner
R1	Cyberattack on city systems	Reduce	Firewalls, 24/7 monitoring, regular security testing	Risk & Quality Manager
R2	Technical issues with smart devices	Reduce	Test devices before rollout and keep spare units	Technical Manager
R3	Budget overrun	Reduce	Monthly cost review; use the EGP 50M backup budget	Finance Manager
R4	Data privacy concerns	Avoid	Clear privacy policy and follow data protection law	Risk & Quality Manager
R6	Contractor or supplier delays	Transfer	Contracts with deadlines and late-delivery penalties	Planning Manager
R7	Insufficient user adoption	Reduce	Awareness campaigns, easy app, training sessions	Stakeholder Manager
R5	Delays in regulatory approvals	Reduce	Start approvals early and assign one person to follow up	Project Manager

5. Contingency Plans (High Risks)

ID	Risk	Trigger	Emergency action
R1	Cyberattack on city systems	An attack or data breach is detected	Isolate the affected systems, activate the incident response team, inform authorities and citizens
R2	Technical issues with smart devices	More than 5% of devices fail in one area	Replace with spare units and switch to manual operation until fixed
R3	Budget overrun	Spending is more than 10% over plan	Release the backup budget and delay low-priority work
R4	Data privacy concerns	Public complaints or legal questions	Pause the related service, review the privacy policy and publish a clear statement

6. Communicate the Plan and Train the Staff

- Share the plan and risk register with the whole team at project kick-off.
- Cybersecurity and data privacy training for all staff before the pilot.
- Emergency drills for the control center team every 3 months.
- Inform stakeholders about high risks in the monthly report.

7. Monitor for New Risks

- **Weekly:** High risks are reviewed in the team meeting.
- **Monthly:** the Risk & Quality Manager updates the risk register and scores.
- **End of each phase:** a full risk review to find new risks for the next phase.
- **Anyone can report a new risk;** it is added to the register within one week.